Data Analytics Internship

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Domain: Data Analytics

Project: Business Insights Dashboard using Superstore Dataset

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PHASE 1 — BLUEPRINT DOCUMENT

Project Title

Business Insights Dashboard using Superstore Dataset

Project Type

Individual Project

1. Problem Statement

Businesses generate large amounts of sales data, but it is difficult to understand key insights from this data. Traditional reports do not clearly show which areas are performing well or where losses are occurring. This makes it challenging for decision-makers to take informed actions.

2. Dataset Details

- **Dataset Name:** Superstore Dataset
- **Type:** Structured CSV data
- **Description:** Contains retail sales data including sales, profit, discount, and customer details

Key Columns:

- Sales
- Profit
- Discount
- Region
- Category and Sub-Category
- Quantity

- Order Date

3. Hypothesis

- High discounts may lead to lower profit
- Profitability varies across different regions
- Some categories contribute more to total sales than others

4. Solution Approach

This project involves analyzing the Superstore dataset using Python to extract meaningful insights and trends. Data cleaning and exploratory data analysis (EDA) will be performed, followed by creating visualizations to understand business performance. The insights will be presented through an interactive Power BI dashboard to support better decision-making.

5. Analysis Plan with Expected Visualizations

- **Overall Performance Analysis**
  → KPI Cards (Total Sales, Total Profit, Total Quantity)
- **Sales by Category**
  → Bar Chart
- **Profit by Region**
  → Bar Chart
- **Sales Trend Over Time**
  → Line Chart
- **Discount vs Profit Analysis**
  → Scatter Plot
- **Top Customers by Sales**
  → Bar Chart

- **Category Contribution to Revenue**
  → Pie / Donut Chart

6. Tech Stack

- **Python (Jupyter Notebook)** – Data cleaning and analysis
- **Pandas, NumPy** – Data processing
- **Matplotlib, Seaborn** – Data visualization
- **Power BI** – Dashboard creation
- **React.js** – Interactive dashboard development

7. Success Criteria

- The dashboard clearly displays key business metrics such as sales, profit, and quantity
- Visualizations provide meaningful and easy-to-understand insights
- The analysis identifies trends, patterns, and loss areas
- The final dashboard is interactive and user-friendly